^{33}S(α,d) **1975Na18,1977Na10**

J^{π}=3/2^{+} for ^{33}S ground state.

1975Na18,1977Na10: a 40-MeV α beam was produced from the Michigan State University cyclotron. Targets were 76.8%-enriched ^{33}S of 15 μg/cm^{2} sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured σ(E$_d$,θ). Deduced levels, J, π, and L-transfers.

^{35}Cl Levels

E(level)†	L†	Comments
6200 *10*	6	
7170 *10*	6	
7670 *10*	6	
7750 *10*	6	
7870 *10*	4+6	L: 1977Na10 states that this is excited by a mixture of L=4 and 6 transfers. The L=6 admixture can be attributed to a $(f_{7/2})^2_{7,0}$ transfer whereas the L=4 component is associated with a mixture of $(f_{7/2})^2_{5,0}$ and $(f_{7/2},p_{3/2})_{5,0}$ transfers.
8010 *10*	6	
8100 *10*	6	
8700 *10*	6	
8840 *10*	6	
9150 *10*	6	
9450 *10*	6	

† From 1977Na10. Experimental angular distributions of L=4, 5, 6 to ^{42}Sc and ^{34}Cl levels with known J^{π} are compared with the σ(E$_d$,θ) measured in 1977Na10 to assign L-transfers.